ReSTIR DI

ReSTIR 1 ray per pixel
BRDF: Lambert + GGX (different roughness)
Init samples 5 x MIS(Area Light Importance, GGX)
Temporal accumulation: 4 frames
Spatial accumulation: 12 samples

ReSTIR Lambert spatial join study in low res i.e. 512×512

BRDF: Lambert
Init samples 5 x Area Light Importance
Temporal accumulation: 2 frames
Spatial accumulation: 5 samples radius 10

no restriction on join

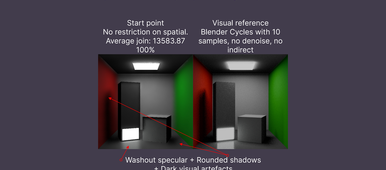
nor's cos >= 0.9
fireflies on box edge removed

nor's cos >= 0.9
pos's dist < 0.04
better shadow
lower convergence

Blender Cycles with 10 rays per pixel, no denoise, no indirect

ReSTIR GGX spatial join study

BRDF: GGX (different roughness)
Init samples 5 x MIS(Area Light Importance, GGX)
Temporal accumulation: 2 frames
Spatial accumulation: 5 samples radius 10



nor's cos < 0.9
pos's dist < 0.04
Specular over spreads

nor's cos >= 0.9
pos's dist < 0.04
half's cos >= 0.9
Adding restriction between half vectors narrows the specular

nor's cos >= 0.9
pos's dist < 0.04
half's cos >= 0.98
better specular
lower convergence

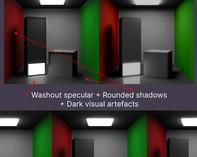
Blender Cycles with 10 rays per pixel, no denoise, no indirect

Average numbers of samples merged per pixel using different join restrictions:
73.51 [100%] - no restriction on spatial join
57.97 [78%] - pos's dist < 0.1
35.07 [47%] - pos's dist < 0.04 [good shadows]
33.12 [45%] - pos's dist < 0.1 and nor's cos >= 0.9 [OK shadows, good lambert]
26.35 [35%] - pos's dist < 0.1 and nor's cos >= 0.9 and half's cos >= 0.98 [ok shadows, good lambert, good ggx]
49.41 [67%] - half's cos >= 0.98

Lambert + GGX, 5 x MIS(Area Lights Importance, GGX), Temporal 4, Spatial 12 from radius 10
Visual impact & reservoir join percentage vs join restriction

Start point
No restriction on spatial.
Average join: 13583.87
100%

Visual reference
Blender Cycles with 10 samples, no denoise, no indirect



Washout specular + Rounded shadows
+ Dark visual artefacts

half's cos >= 0.98
average join: 6768.61
49.82%
+ Good specular

half's cos >= 0.98
nor's cos > 0.9
average join: 6478.03
47.68%

half's cos >= 0.98
nor's cos > 0.9
pos's dist < 0.5
average join: 6469.80
47.62%
+ No artefacts

half's cos >= 0.98
nor's cos > 0.9
pos's dist < 0.1
average join: 3692.08
27.17%
+ Ok, shadows

half's cos >= 0.98
nor's cos > 0.9
pos's dist < 0.04
average join: 190.19
1.40%
+ Good Shadows

- Artefacts: nor's cos >= 0.9 and pos's dist < 0.5 is needed to remove artefacts

- Lambert: It looks good everywhere (once the artefacts are removed)

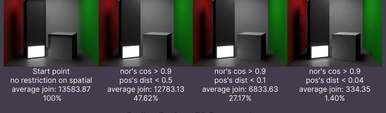
- GGX: half's cos >= 0.98 results in good specular but we drop half the spatial samples

- Shadows: best shadows with pos's dist < 0.04 (worse joining 1.4%). Acceptable shadows begin from pos's dist < 0.1, 27.17%

A lower percentage of considered neighbors joining means that convergence will require more frames.

Lambert + GGX, 5 x MIS(Area Lights Importance, GGX), Temporal 4, Spatial 12 samples no half's restriction

Radius 10 - Area 314 reservoirs



Start point
no restriction on spatial
average join: 13583.87
100%

nor's cos > 0.9
pos's dist < 0.5
average join: 12783.13
47.62%

nor's cos > 0.9
pos's dist < 0.1
average join: 6833.63
47.62%

nor's cos > 0.9
pos's dist < 0.04
average join: 334.35
1.40%

Radius 5 - Area 78.5 reservoirs

Start point
no restriction on spatial
average join: 14521.27
100%

nor's cos > 0.9
pos's dist < 0.5
average join: 14092.63
97.04%

nor's cos > 0.9
pos's dist < 0.1
average join: 13095.53
90.18%

nor's cos > 0.9
pos's dist < 0.04
average join: 5553.23
38.24%

Radius 2.5 - Area 19.625 reservoirs

Start point
no restriction on spatial
average join: 15011.42
100%

nor's cos > 0.9
pos's dist < 0.5
average join: 14743.55
98.21%

nor's cos > 0.9
pos's dist < 0.1
average join: 14698.51
97.91%

nor's cos > 0.9
pos's dist < 0.04
average join: 11608.62
77.33%

Ignoring GGX half's cos and gradually concentrating area results in better joining percentage, better specular and better shadows but it gradually adds fireflies. Oh, boy wouldn't it be nice if we can join shadow separately?

Lambert + GGX, 5 x MIS(Area Lights Importance, GGX), Temporal 4, Spatial 12 samples small disk area

12 samples uniformly distributed over radius 10
Occasionally collision.

12 samples uniformly distributed over radius 5
There's collision.

12 samples uniformly distributed over radius 2.5
Collision city.

For small disk areas instead of computing the sample using r = √(rand), angle = rand * 2 * π we could accept disk samples based on the probability p=num_samples/disk_area_samples. On average 12 disk samples will be generated without collision. This implementation, while it offered good specular for half's cos > 0.9 still has fireflies.

radius 10

half's cos >= 0.98
nor's cos > 0.9
pos's dist < 0.1
average join: 3692.08
27.17%

radius 3 - probabilistic samples
half's cos >= 0.9
nor's cos > 0.9
pos's dist < 0.1
average join: 11420.60
67.68%

radius 3 - probabilistic samples

No collision.
Fireflies.

In the disk sampler instead of using r = √(rand) we could use r = rand ** factor and a factor of 1 or 2 would bias samples closer to the center. Changing the distribution requires integration of pdf in reservoir joining for unbiased results.

With factor 1,2 specular is acceptable even with half's cos >= 0.9, shadows are improved and fireflies have been reduced.

radius 10 factor 2

Collisions.
Reduced fireflies

radius 10 factor 2

half's cos >= 0.9
nor's cos > 0.9
pos's dist < 0.1
average join: 7526.37
52.38%

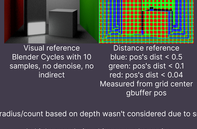
radius 10 factor 1/2 (uniform reference)
half's cos >= 0.98
nor's cos > 0.9
pos's dist < 0.1
average join: 3692.08
27.17%

radius 10 factor 1

half's cos >= 0.9
nor's cos > 0.9
pos's dist < 0.1
average join: 6049.12
42.38%

radius 10 factor 1

Less collisions.
Even less fireflies.



Visual reference
Blender Cycles with 10 samples, no denoise, no indirect

Distance reference
blue: pos's dist < 0.5
green: pos's dist < 0.1
red: pos's dist < 0.04
Measured from grid center
gbuffer pos

Reducing the sampling radius/count based on depth wasn't considered due to small scene depth range.

In higher resolution things are a lot easier.